

MAURICE BISHOP INTL, GrenadaWMO: **789580**Lat: **12.004N**Lon: **61.786W**Elev: **7**StdP: **101.24**Time Zone: **-4.00 (W04)**Period: **94-19**WBAN: **00409****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.9	23.3	19.5	14.3	25.5	20.0	14.7	25.6	11.1	28.1	10.4	28.0	2.2	60	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	4.8	31.4	26.9	31.2	26.8	31.0	26.8	27.8	30.3	27.5	30.1	27.2	29.9	6.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.8	30.1	26.8	22.4	29.8	26.2	21.7	29.4	88.8	30.7	87.2	30.4	86.0	30.7	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.6	9.1	DB	21.0	33.1	1.8	1.6	19.7	34.2	18.7	35.1	17.7	36.0	16.4	37.2	(4)
(5)			WB	19.5	28.8	1.5	0.7	18.5	29.3	17.6	29.7	16.7	30.0	15.7	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.5	26.6	26.9	27.7	28.1	28.0	27.9	28.1	28.5	28.2	27.7	27.1
	DBStd	1.00	0.73	0.74	0.80	0.77	0.69	0.81	0.71	0.77	0.87	0.87	0.81	0.75
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6426	513	464	526	529	562	539	554	561	555	563	531	531
	CDD18.3	3384	254	231	267	279	303	289	296	302	305	304	281	272
	CDH23.3	35768	2200	1994	2497	2924	3419	3218	3276	3485	3639	3484	3004	2627
	CDH26.7	10219	439	394	554	773	1028	932	954	1146	1344	1169	868	617
Wind	WSAvg	5.3	5.1	5.9	6.0	6.3	6.5	6.4	5.3	4.5	4.6	4.5	4.5	4.8
Precipitation	PrecAvg	1602	108	58	49	60	86	166	170	187	163	183	230	143
	PrecMax	2002	353	156	108	265	236	283	272	254	328	365	498	452
	PrecMin	1161	39	10	8	14	25	78	89	114	72	75	87	41
	PrecStd	171	64	33	27	55	61	57	41	40	59	68	91	90
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	30.1	30.2	31.1	31.2	31.2	31.2	31.9	32.2	32.0	31.2	30.9
		MCWB	24.8	24.7	25.1	26.3	26.5	26.3	26.5	27.0	27.3	27.1	26.8	26.0
	2%	DB	29.7	29.6	30.0	30.2	30.9	30.8	30.9	31.2	31.6	31.2	31.0	30.2
		MCWB	24.7	24.5	25.1	25.7	26.4	26.3	26.5	26.9	27.1	27.0	26.7	25.7
	5%	DB	29.2	29.1	29.4	30.1	30.2	30.2	30.2	30.9	31.1	31.0	30.2	29.9
		MCWB	24.6	24.4	24.7	25.6	26.0	26.1	26.2	26.9	27.0	27.0	26.4	25.6
	10%	DB	28.9	28.9	29.1	29.8	30.0	30.0	30.0	30.2	30.9	30.5	30.0	29.2
		MCWB	24.5	24.3	24.6	25.4	25.9	26.0	26.1	26.5	26.9	26.8	26.3	25.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.3	26.6	27.2	27.7	27.6	27.5	28.1	28.2	28.1	27.8	27.1
		MCDB	28.8	28.7	29.2	29.9	30.0	29.8	29.8	30.4	30.8	30.6	30.1	29.6
	2%	WB	25.7	25.7	26.0	26.7	27.1	27.0	27.1	27.4	27.7	27.6	27.2	26.6
		MCDB	28.5	28.5	28.9	29.5	29.6	29.5	29.6	30.0	30.5	30.3	29.8	29.2
	5%	WB	25.3	25.2	25.6	26.2	26.7	26.7	26.7	27.1	27.3	27.2	26.9	26.2
		MCDB	28.2	28.2	28.6	29.1	29.2	29.2	29.3	29.8	30.2	29.9	29.6	28.9
	10%	WB	24.9	24.7	25.2	25.8	26.4	26.4	26.5	26.8	27.1	27.0	26.6	25.7
		MCDB	27.9	27.8	28.2	28.8	29.0	28.9	29.1	29.6	30.0	29.7	29.2	28.5
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.0	4.8	4.4	4.1	4.1	4.5	4.7	4.8	4.9	4.9	5.0
		MCDBR	5.3	5.2	5.0	4.5	4.3	4.4	4.7	5.1	5.1	5.1	5.2	5.3
		MCWBR	2.7	2.7	2.6	2.4	2.3	2.3	2.4	2.7	2.6	2.7	2.7	2.8
	5% WB	MCDBR	4.9	4.8	4.6	4.2	4.1	4.3	4.5	4.8	4.8	4.9	4.9	5.0
		MCWBR	2.8	2.9	2.7	2.5	2.4	2.4	2.6	2.8	2.7	2.8	2.8	2.9
Clear-Sky Solar Irradiance	taub	0.379	0.398	0.399	0.441	0.470	0.531	0.500	0.442	0.420	0.389	0.381	0.388	
	taud	2.585	2.506	2.516	2.404	2.309	2.108	2.220	2.447	2.502	2.621	2.633	2.584	
	Ebn at Noon	926	918	919	871	831	772	800	859	886	915	918	908	
	Edh at Noon	98	109	110	122	132	159	143	116	110	96	93	96	
All-Sky Solar Radiation	RadAvg	5.42	6.08	6.48	6.63	6.43	5.96	6.13	6.30	6.24	5.67	5.18	5.09	
	RadStd	0.21	0.20	0.28	0.25	0.39	0.31	0.29	0.22	0.24	0.26	0.36	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.20	+0.33	+0.41	+0.14	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page